

KIDS' PLAYROOM BUILD

Corner Loft · Ranger's Outpost Playhouse · bridge-ready

CONSTRUCTION DOCUMENT SET · dimensional lumber + stock hardware · generated from the parametric Blender model

ROOM 13'-6" x 13'-5" CEILING 8'-0" PLATFORM 3'-8" GUARD 3'-0" GUARD OPENINGS 2.59"-2.70"

DESIGNED FOR 18 MONTHS THROUGH ABOUT 8 YEARS — IN THREE PHASES

READ FIRST — FIVE THINGS IN THE BRIEF THAT DO NOT WORK AS WRITTEN

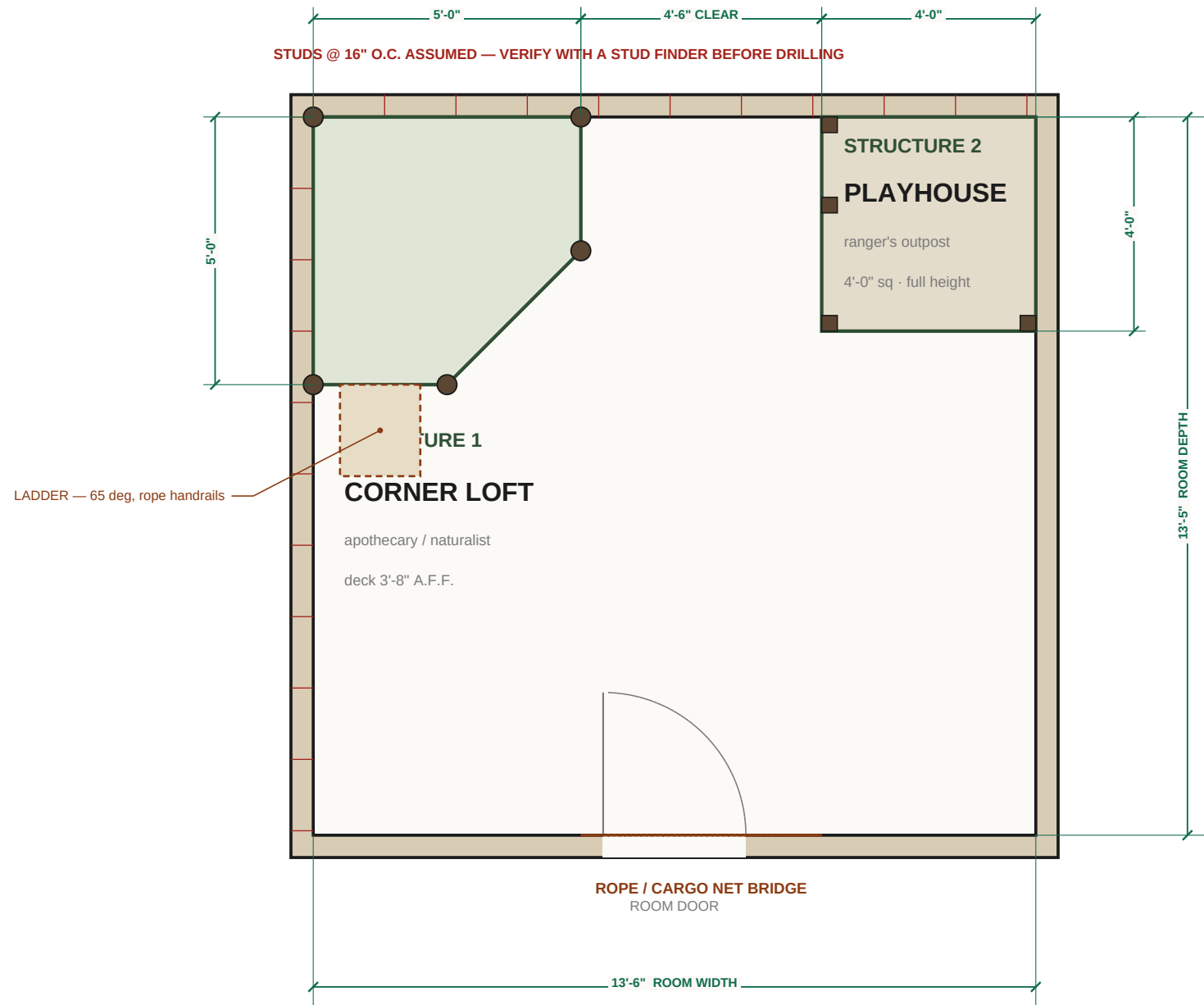
1. This has to work for an 18-month-old AND a four-year-old. That drove a redesign. BRIEF one structure, one age CPSC caps a TODDLER platform (6-23 months) at 32". This deck is 44". You cannot make a 44" loft safe for an 18-month-old, so the loft is not his. The enclosed ground den and the playhouse are, from day one. The ladder LIFTS OFF, which is what actually keeps him out of the loft. Full phase table on this sheet.	4. The knee wall studs are 34-3/4", not 40-42" BRIEF studs ~40-42" tall Same arithmetic. The wall dies into the underside of the joists at 37-3/4", and two plates eat 3" of that. Cut one, hold it in place, and confirm before you cut the other fifteen.
2. The rope net guardrail had to go. It is a head-entrapment hazard. BRIEF rope net railing, 4-6" mesh CPSC: any opening between 3.5" and 9" lets a small body through and catches the head. A 4-6" mesh fails both probes. Vertical balusters at under 3-3/8" clear instead, with a 2x4 toe board closing the gap at deck level. Rope stays as tight decorative wrapping on the posts. No slack, no loops: a loose loop is a strangulation risk under 3.	5. The bridge span is 54", not 42" BRIEF span ~42" THIS SET clear span 54" 162" wall - 60" loft - 48" playhouse = 54" of clear air between them. Both structures are in corners, so neither can slide toward the other to close the gap. Order the net for 54". Expect 6-8" of sag at that span, which is why the plan set carries a top rope AND a mid rope.
3. A 36-38" hobbit door will not fit under a 44" platform BRIEF R.O. 24-26" W x 36-38" H THIS SET R.O. 26" W x 32" H 44" deck - 3/4" plywood - 5-1/2" joist = 37-3/4" of wall, floor to underside of framing. Take out the 1-1/2" bottom plate, a 1-1/2" flat header, a 2-3/4" cripple and a 1-1/2" top plate and 32" is what is left. A 32" door is still right for a 3-6 year old. The semicircular arch on top reads taller than it measures.	THE THREE PHASES 18 mo - 3 yr Ground den + playhouse. Ladder stored OFF the structure. Nothing to fall from. Both spaces are at floor level. 3 yr + Hang the ladder. Loft in use. Gate self-closes. Only once he climbs it unassisted, both ways, every time. 5 yr + (or when the youngest is 4) Unscrew the panel, bolt the eye bolts, hang the net. Two hours. Every provision is already in the wall.

SHEET INDEX A0 Cover · critical notes · phases A1 Overall room plan · anchor zones · studs A2 Loft deck framing · hexagon geometry A3 Loft developed elevation · guards · gate A4 Knee wall · hobbit door · arch template A5 Removable ladder · hook + pin · den shelf A6 Playhouse plan · elevation · roof framing A7 Bridge — DEFERRED. Provisions + load check A8 Cut list A9 Hardware · Home Depot buy list A10 Reference model views A11 Technical orthographic views	AT A GLANCE Deck 22.36 sq ft hexagonal, 60" x 60" envelope Angles 90/90/157.5/135/157.5/90 — miters 11.25 & 22.5 Posts 6 at the loft, 4 at the playhouse Guards 2x2 balusters, 2.59"-2.70" clear, 36" tall Toe board 2x4 on edge — no gap at deck level Ladder 5 rungs @ 7-5/16" — lifts off, 2 pins Gate self-closing, swings in, no lock Bridge deferred — provisions in the wall now Lumber ~65 lf 4x4 · ~137 lf 2x4 · ~55 lf 2x2 Plywood 4 sheets 3/4" Cost \$900 - \$1,150 phase 1 (sheet A9) Build 2 weekends, one person and a helper	WHAT THIS SET IS MEASURED AGAINST CPSC Public Playground Safety Handbook, Pub. 325 · 3.5" torso probe — no guard opening passes it · 3.5"-9" head entrapment window avoided throughout · toddler platform cap of 32" — why the loft is 3 yr+ · protrusion limit — cap nuts on every through-bolt ASTM F1148, home playground equipment 16 CFR 1213 bunk-bed guardrail gap rule, 3.5" NOT ENGINEER-STAMPED AND NOT CERTIFIED. Designed to these rules, tested against none of them. Verify stud spacing on site before drilling a ledger. Nothing here replaces watching a two-year-old.
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OVERALL ROOM PLAN

structure locations · wall anchor zones · stud layout

A1



LOAD PATH

The wall ledgers carry the real load.
Posts are secondary / aesthetic.

LOFT: two 2x6 PT ledgers, one on the north wall and one on the west wall, 3/8" x 4" lag + washer into EVERY stud.

PLAYHOUSE: north and east posts lag-bolted directly to wall studs.

BRIDGE: both ends through-bolted into 2x6 ledgers that are themselves lagged into a minimum of 2 studs.
Eye BOLTS with washer + nylock nut.
Never screw eyes.

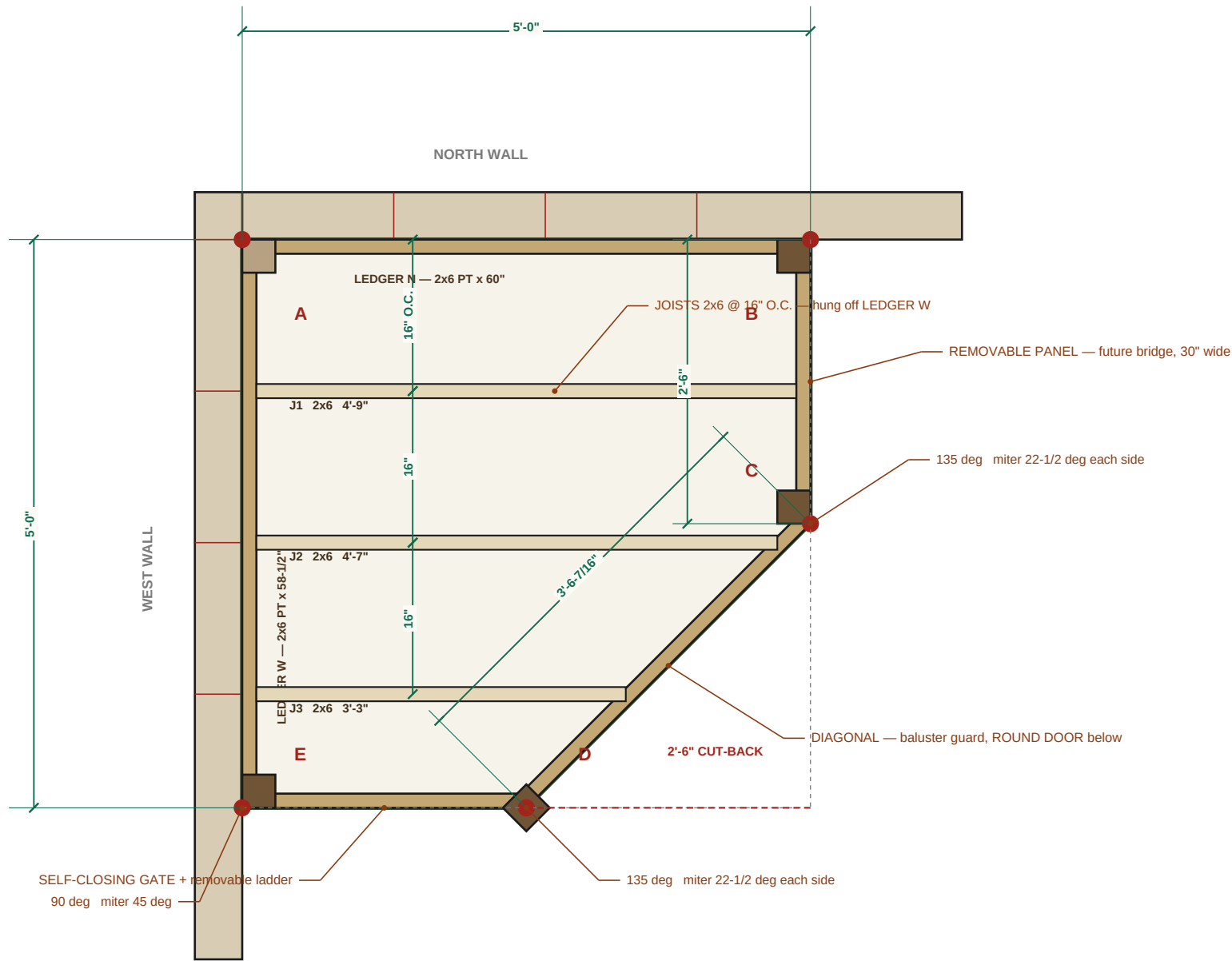
VERIFY STUD SPACING ON SITE.

16" O.C. is assumed throughout; if your studs are 24" O.C. or irregular, shift ledger bolt locations to land on wood and add a bolt — do not rely on drywall anchors anywhere in this set.

LOFT — DECK FRAMING PLAN

pentagon geometry · joist layout · post schedule

A2



PENTAGON GEOMETRY

VERTEX	u (in)	v (in)	POST CUT
A	0.000	0.000	4x4 x 37-3/4"
B	60.000	0.000	4x4 x 80"
C	60.000	30.000	4x4 x 80"
D	30.000	60.000	4x4 x 80"
E	0.000	60.000	4x4 x 80"

WHY THIS SHAPE

Three exposed faces: straight off the north wall, one 45-degree diagonal across the corner, straight off the west wall. Interior angles 90/90/135/135/90.

That gives exactly TWO saw settings for the whole deck: 45 degrees where a rim meets a wall, and 22-1/2 degrees at the two diagonal corners. Nothing else in this build needs an angle cut.

FACES 30" / 42-7/16" / 30"

DECK AREA 21.88 SQ FT

POSTS 4 tall + 1 corner stub

Both 60-inch wall runs stay full length, so both ledgers get maximum stud engagement.

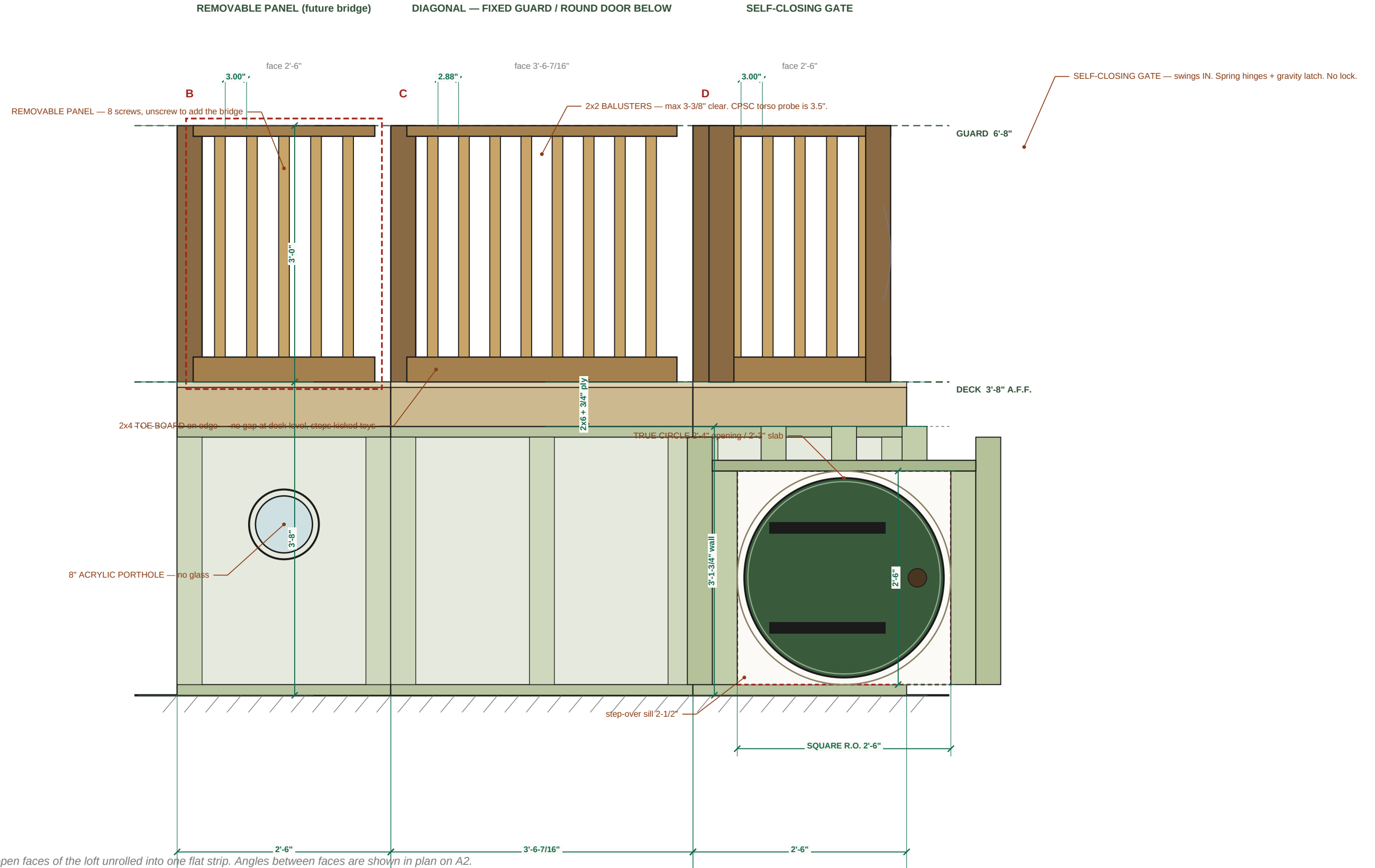
JOIST NOTES

J1 is square both ends - Simpson LUS26 hangers. J2 and J3 die into the diagonal at 45 degrees. Use Simpson LSU26 adjustable skewable hangers, bent to the angle. Two short blocks behind the diagonal carry the deck edge. Do not toe-nail either one.

LOFT — DEVELOPED ELEVATION

the three open faces unrolled flat

A3

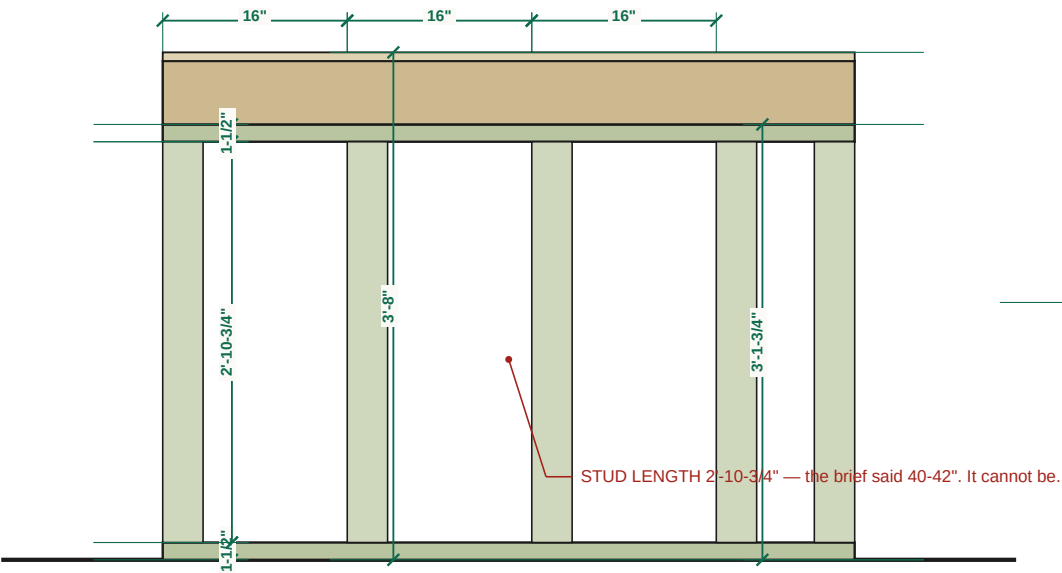


LOFT — KNEE WALL + HOBBIT DOOR

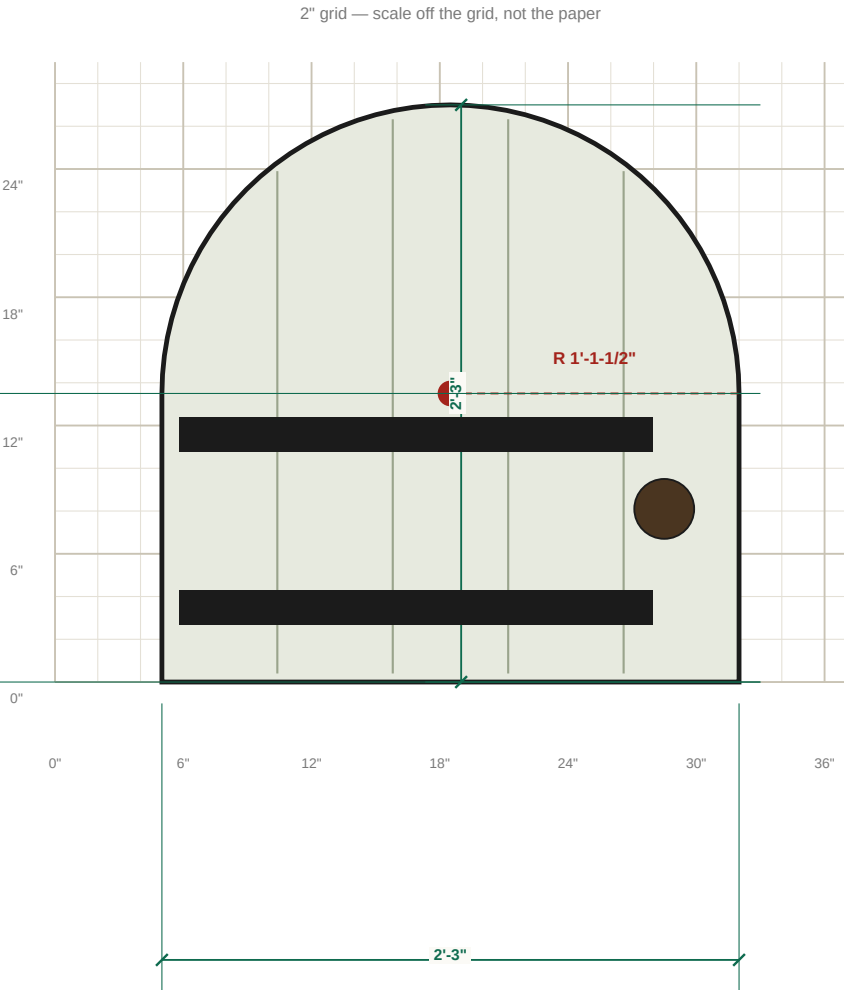
wall section · full arch template on a 2" grid

A4

KNEE WALL SECTION — typical



HOBBIT DOOR — FULL TEMPLATE



DOOR BUILD-UP

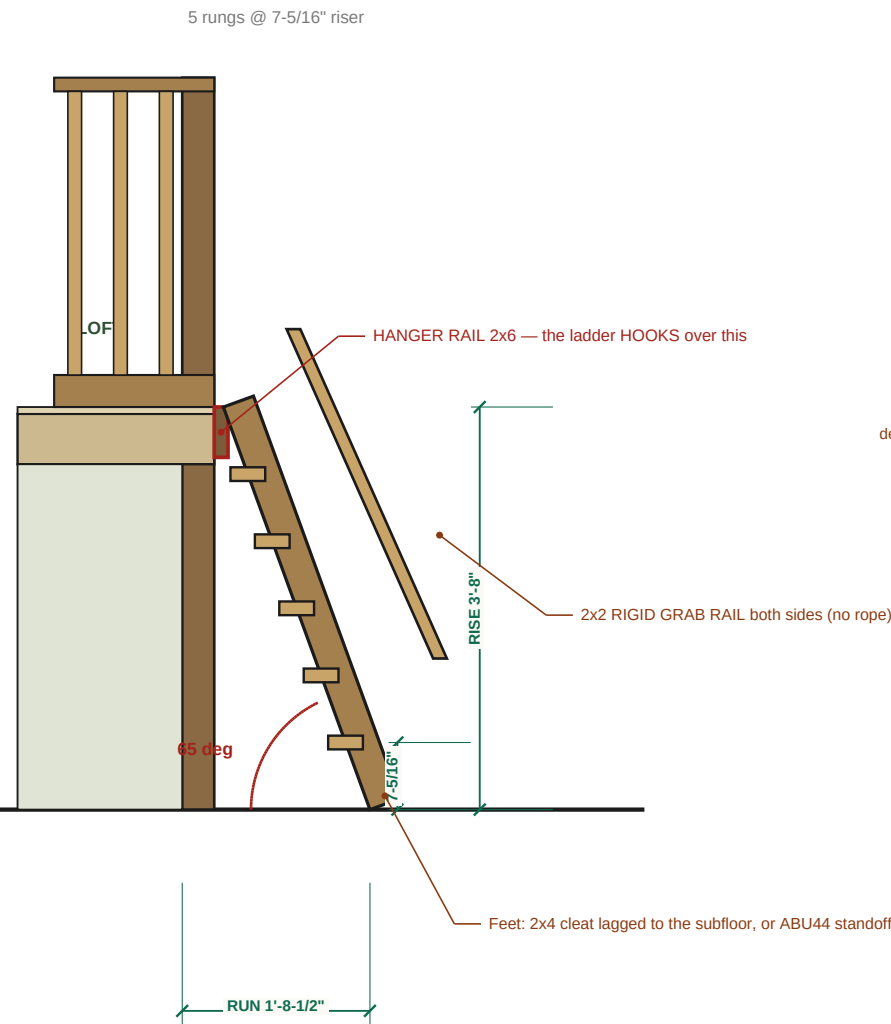
- Slab: two layers of 3/4" ply, glued and screwed, grain crossed, cut to the template above = 1-1/2" thick.
- Face with 1x4 tongue-and-groove pine, vertical, ripped to the arch with a jigsaw.
- Hinges: 12" black strap hinges, 2 per door, through-bolted with 1/4" carriage bolts — screws alone will sag.
- Jamb: 3/4" ply, 3 sides, glued and screwed into the king/jack studs. No sill — remove the bottom plate inside the R.O. after the wall is standing.
- Knob: 2-3/4" round wood knob, 1/4" lag from the back.
- Magnetic catch, not a latch. A kid must be able to push the door open from the inside with no hardware.
- NO lock of any kind, inside or out.
- Paint: deep green (see A9). Two coats, sand between.

REMOVABLE LADDER + DEN SHELF

elevation · hook and pin detail · ground-level shelf

A5

LADDER ELEVATION — LIFTS OFF

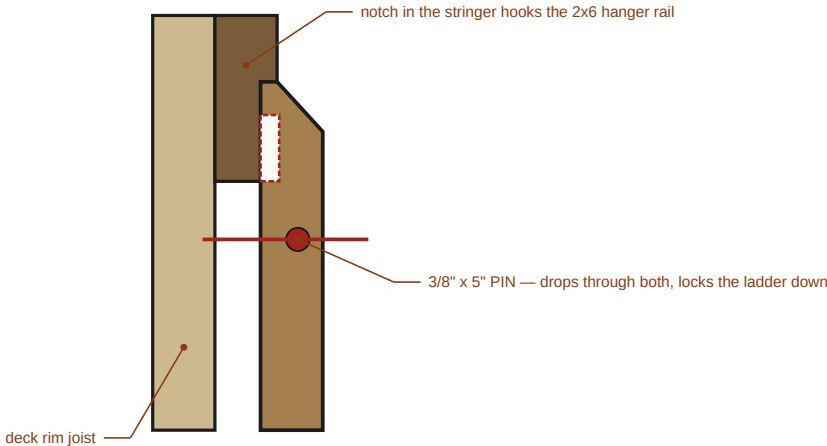


WHY THE LADDER LIFTS OFF

A gate stops a fall. It does not stop an 18-month-old, because whatever latch a 4-year-old can work, a determined toddler eventually works too.

The ladder is the real access control. Pull two pins, lift it off, stand it behind the playhouse. The loft is then physically unreachable — there is nothing to climb.

Hang it when the 4-year-old is up there. Pull it at night and any time the little one is loose in the room.

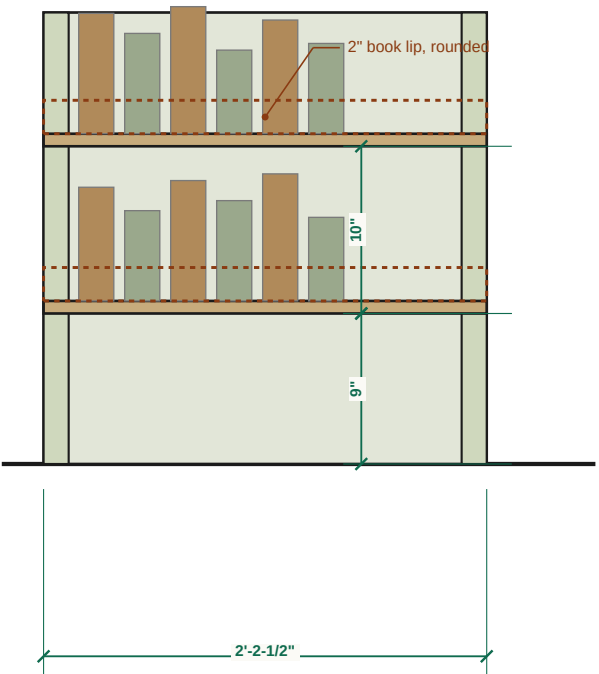


HOOK + PIN DETAIL

pull two pins, lift the ladder off, store it flat

DEN SHELF — GROUND LEVEL

inside the enclosed base, NOT in the guardrail



WHY THE SHELVES CAME OUT OF THE RAILING

You were right, and for a better reason than you think. A shelf bolted into a guardrail is a ladder. A 4-year-old puts a foot on the bottom shelf and their centre of mass clears a 36" rail. The guard stops working.

The books and the apothecary jars move to floor level, inside the den, where the 18-month-old actually plays.

CHOKING — READ THIS

Acorns, dried botanicals and anything under 1-3/4" is a choking hazard until 3. Either hot-glue the jar lids shut so they are display-only, or leave the jars out until the little one is past it. Board books only down here.

CUT LIST

every member, every finished length

A6

CUT LIST — generated directly from the model. Every length is the finished cut length.

Lengths are long-point to long-point where a member is mitered. Cut one of any repeated part, test-fit it, then batch the rest.

LOFT / POSTS

STOCK	LENGTH	QTY	NOTE
4x4 PT	3'-1-3/4"	1	room-corner stub
4x4 PT	6'-8"	1	bridge post
4x4 PT	6'-8"	3	

LOFT / LEDGERS

STOCK	LENGTH	QTY	NOTE
2x6 PT	5'-0"	1	lag to studs @16 O.C., 3/8 x 4 lag + washer
2x6 PT	4'-10-1/2"	1	lag to studs @16 O.C.

LOFT / RIM

STOCK	LENGTH	QTY	NOTE
2x6	2'-6"	2	outside face length 30.000"
2x6	3'-6-7/16"	1	outside face length 42.426"

LOFT / JOISTS

STOCK	LENGTH	QTY	NOTE
2x6	4'-9"	1	LUS26 hanger both ends
2x6	4'-7"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	3'-3"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	2'-1-7/16"	2	decking support behind the diagonal rim, 45 deg both

LOFT / KNEE WALL

STOCK	LENGTH	QTY	NOTE
2x4 PT	2'-6"	2	30.00" face length
2x4	2'-6"	2	30.00" face length
2x4	2'-10-3/4"	7	34.75" - NOT 40-42"
2x4 PT	3'-6-7/16"	1	42.43" face length
2x4	3'-6-7/16"	1	42.43" face length

LOFT / ROUND DOOR

STOCK	LENGTH	QTY	NOTE
2x4	2'-6"	2	30" jack
2x4	2'-10-3/4"	2	34.75" king
2x4	3'-1"	1	flat - deck load bypasses this wall
2x4	4-3/4"	3	4.75"
3/4" ply	8"	4	router-cut quadrant; four make the 28" circle inside

LADDER

STOCK	LENGTH	QTY	NOTE
2x6	2'-0"	1	lag to rim + both posts; ladder hooks over this, 3/8
2x4	4'-0-9/16"	2	48.55" o/a; notch top to hook the hanger rail, 25 de
2x2	3'-5-7/16"	2	1/2" roundover, sand 180. RIGID - sagging rope is a
2x2	3-3/4"	4	grab-rail standoff block
2x4	1'-3"	5	riser 7.33"; dado 3/4" into stringers + 2 screws/end

LOFT / GUARDS

STOCK	LENGTH	QTY	NOTE
2x4	2'-1-1/2"	1	REMOVABLE - 8 screws. Take it off to add the bridge
2x4	2'-1-1/2"	2	on edge at deck level; closes the bottom gap, stops
2x2	2'-7"	10	3.000" clear gap - CPSC limit is 3.5"; 1/4" roundove
2x4	3'-1-15/16"	1	fixed guard; top of rail = 80" = 36" above deck
2x4	3'-1-15/16"	1	on edge at deck level; closes the bottom gap, stops
2x2	2'-7"	8	2.881" clear gap - CPSC limit is 3.5"; 1/4" roundove
2x4	2'-1-1/2"	1	SELF-CLOSING GATE leaf - hinged at Cp, swings IN ove
2x4	3'-0"	2	self-closing spring hinges (pr) / gravity latch + ma

LOFT / DEN SHELF

STOCK	LENGTH	QTY	NOTE
3/4" ply	1'-11-1/2"	2	23.5" x 7.25"; screw to studs, 1/4" roundover, 2" li
1x3	1'-11-1/2"	2	book lip, rounded

SHEET GOODS + WHAT TO ACTUALLY BUY

ITEM	NET	BUY
4x4 x 8 ft PT post	29.8 lf	8
4x4 x 10 ft PT post	(1 post + the stub)	1
2x6 x 8 ft Ground Contact PT	9.9 lf	3
2x6 x 8 ft SPF	27.4 lf	5
2x4 x 8 ft Ground Contact PT	8.5 lf	2
2x4 x 8 ft SPF stud	79.0 lf	18
1x3 x 8 ft pine (book lips)	3.9 lf	1
3/4" 4x8 ply — deck + shelf	21.75 sq ft + shelf	2
3/4" 4x8 ext ply — skins + doors	~46 sq ft	2
1x4 T&G pine — door faces	~24 lf	1 bdl

2x quantities include 15% waste. Every cut in this set fits an 8 ft stick.
The one 10 ft 4x4 yields an 80" post AND the 37-3/4" corner stub with almost no waste. Nine 8-footers would leave you one post short.

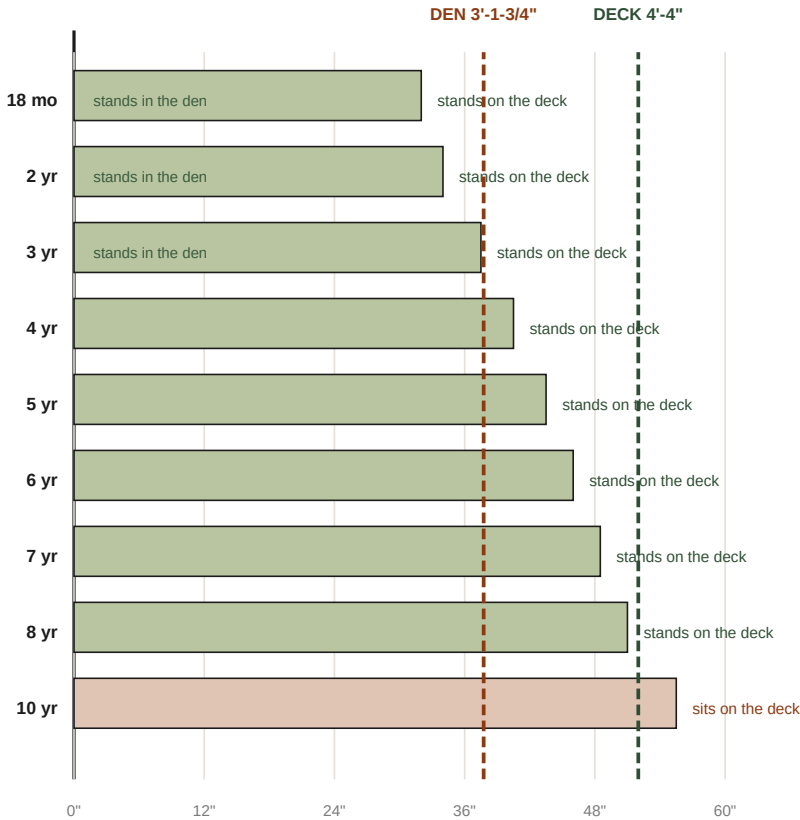
HARDWARE SCHEDULE + HOME DEPOT BUY LIST

Internet # verified against homedepot.com. Prices are US national ranges and move weekly — treat them as a budget, not a quote. Items without an Internet # are commodity stock carried in every store; grab them off the shelf.

ITEM	INTERNET #	QTY	EST. EACH	WHAT IT DOES / WHY THIS ONE
PHASE 1 LUMBER				
4 in. x 4 in. x 8 ft. #1 Pressure-Treated Post	202812131	8	\$12 - \$16	four loft posts at 80" + four playhouse posts at 84"
4 in. x 4 in. x 10 ft. #1 Pressure-Treated Post	—	1	\$16 - \$22	fifth 80" post AND the 37-3/4" corner stub, almost no waste
2 in. x 6 in. x 8 ft. Ground Contact PT	—	3	\$9 - \$13	loft ledgers + BOTH future-bridge ledgers (install now)
2 in. x 6 in. x 8 ft. Prime Kiln-Dried SPF	—	6	\$7 - \$10	joists, rim, ladder hanger rail, playhouse header, ridge
2 in. x 4 in. x 8 ft. Ground Contact PT	—	2	\$6 - \$9	knee wall bottom plates (they sit on the subfloor)
2 in. x 4 in. x 8 ft. Prime Kiln-Dried Stud	—	20	\$4 - \$6	studs, plates, rafters, ladder, top rails, toe boards
2 in. x 2 in. x 8 ft. Furring / S4S Pine	—	9	\$3 - \$5	GUARD BALUSTERS + ladder grab rails — ~55 lf
1 in. x 3 in. x 8 ft. Pine Board	—	1	\$6 - \$9	den shelf book lips
3/4 in. 4x8 Sanded Plywood (BC or Sande)	—	4	\$48 - \$68	2 structural, 2 skin + doors + den shelf
1 in. x 4 in. Tongue & Groove Pine (bundle)	—	1	\$28 - \$40	door faces, both doors
STRUCTURAL HARDWARE				
Simpson Strong-Tie LUS26 Face-Mount Joist Hanger	100374889	8	\$1.77 ea	square joist-to-ledger connections
Simpson Strong-Tie LSU26 Adjustable Skewable U Hanger	100375129	6	\$4 - \$6 ea	J2/J3 into the angled rims — THE part for this
Simpson Strong-Tie ABU44Z Adjustable Standoff Post Base	100375358	3	\$14 - \$19 ea	free-standing posts: loft D, playhouse SW, ladder feet
Simpson Strong-Tie A23 / A35 Framing Angle	—	16	\$0.80 ea	post-to-rim and plate ties
Everbilt 3/8 in. x 4 in. Hex Lag Screw (bulk)	—	60	\$0.55 ea	ledgers into studs, posts into studs
Everbilt 3/8 in. Flat Washer (bulk)	—	100	\$0.15 ea	every lag and every bolt, both sides
Everbilt 1/4 in. x 3 in. Hex Lag Screw	—	40	\$0.40 ea	guards to posts, hinges, hanger rail, trim
Everbilt 3/8 in. Acorn / Cap Nut	—	20	\$0.45 ea	CPSC protrusion rule — cap every bolt end a kid can hit
#9 x 3 in. Star Drive Construction Screw, 5 lb box	—	1	\$34 - \$44	general framing
#8 x 1-5/8 in. Construction Screw, 1 lb box	—	2	\$9 - \$12	decking, skins, balusters — SCREWED, never nailed
LADDER + GATE				
Everbilt 3/8 in. x 5 in. Clevis / Hitch Pin w/ lanyard	—	2	\$4 - \$7 ea	lock the ladder to its hanger rail. Pull 2, lift it off.
Everbilt 3 in. Self-Closing Spring Hinge (pair)	—	1	\$12 - \$18	the deck gate MUST close itself. Do not use plain hinges.
Everbilt Gravity / Gate Latch, black	—	1	\$8 - \$14	NO lock, NO keyed hardware, nothing a kid can shut on themselves
Everbilt Magnetic Push Catch	—	3	\$4 - \$6	both hobbit doors + the gate. Kid must push out from inside.
ROPE — DECORATIVE ONLY NOW				
Everbilt 1/2 in. x 50 ft. Twisted Poly Rope (manila look)	—	1	\$22 - \$32	TIGHT post wraps and rail whipping. No slack, no hanging loops.
Hot glue sticks / exterior construction adhesive	—	1	\$6 - \$10	bed every rope wrap so it cannot be unwound into a loop
DOORS, FINISH, LIGHT				
NUVO IRON 12 in. Black Steel Strap Hinges Kit (2-pack)	320202663	2	\$18 - \$26	one 2-pack per hobbit door
2-3/4 in. Round Wood Cabinet Knob	—	2	\$4 - \$7	hobbit door knobs
8 in. Clear Acrylic Circle / Plexiglass sheet to cut	—	1	\$12 - \$22	porthole — acrylic only, never glass
12 in. Clear Acrylic Circle	—	1	\$16 - \$28	playhouse round window
Commercial Electric LED Puck Light, battery, 3-pack	—	2	\$18 - \$26	2 in the den, 1 in the playhouse. Batteries behind a screwed cover.
BEHR Premium Plus Interior Eggshell — deep green	—	1 qt	\$20 - \$26	match "Dark Everglade" / "Pine Mountain"
Varathane Dark Walnut Wood Stain, qt	—	1	\$14 - \$19	rough-sawn look on the 4x4s and trim
Minwax Polycrylic Clear Satin, qt	—	1	\$20 - \$26	water-based — low odor, and it is a kid's room
Zero-VOC primer, qt	—	1	\$16 - \$22	everything a kid touches, and they mouth everything at 18 months
FALL PROTECTION — NOT OPTIONAL				
2 in. Folding Gymnastics / Tumbling Mat, 4 x 8 ft	—	1	\$90 - \$150	A 44" fall needs ~2". 3/4" foam tiles attenuate about 2 ft — NOT enough.
3/4 in. Interlocking Foam Mat Tiles, 24x24, 6-pack	—	2	\$26 - \$38	general floor, and under the playhouse door threshold

WILL THIS STILL BE FUN IN FIVE YEARS? — fit by age, and the things that only show up after you live with it

Heights are CDC 50th percentile standing height. Your kid will not be average; add or subtract and re-read the table.



STANDING HEADROOM vs AGE

WHAT THE CHART SAYS

- The DEN is a stand-up room only until about 3. After that it is a sit-down cave, which is exactly what a 5-year-old wants it to be. It never stops working; it just changes job.
- The DECK takes a standing child to about 8. Past that they sit up there, which is still where kids read and hide at that age.
- An adult can never stand on the deck. That is deliberate. If you could, they would want you up there, and 22 sq ft is a two-kid deck.

THE HONEST LIFESPAN: 18 MONTHS TO ROUGHLY 9 OR 10.

The bridge in phase 3 is what buys the last three years of it.

ADULT ACCESS — THE ONE PEOPLE FORGET

If a two-year-old bumps their head inside the den, you have to get to them. The hobbit door is 24-1/2" x 31-1/4" finished. An adult CAN crawl it, barely. So: screw the B-C knee-wall skin panel on, never glue and never nail it. Eight screws and the whole panel comes off in about a minute. Mark the screw heads with a dab of paint so you can find them in a hurry.

AIR

Plywood floor above, plywood skin around, one sealed acrylic porthole. Drill a pattern of 1-1/4" holes near the top of two panels. Read as a design detail, and they vent the box. Use 1-1/4", never 3/8" to 1" — that range is a finger-entrapment size.

SPLINTERS BEAT AESTHETICS AT THIS AGE

Rough-sawn is the look you want and the worst possible surface for an 18-month-old who mouths everything. Wire-brush and stain only what nobody grips: posts ABOVE the deck, roof trim, the gable. Sand every graspable surface to 150-180 — balusters, top rails, rungs, grab rails, door edges, the whole den interior. Run a bare forearm over every surface below 40" before you let him in.

STORING THE LADDER

Lay it FLAT behind the playhouse or strap it to a wall. A ladder leaning against a wall is the single most tippable object in the room, and a toddler will pull on it precisely because it is new and out of place.

NO CORDS. NONE.

Battery puck lights only, with the battery cover screwed shut. No plug-in string lights, no extension cords, no lamp on the deck. A cord at 44" is a strangulation hazard and a trip hazard at once.

RE-CHECK SCHEDULE — write the date on the ledger in pencil

30 days: re-torque every lag. Green lumber shrinks and they WILL be loose. Every 6 months: lags again, gate closes and latches, ladder pins present, no baluster gap grown past 3-1/2", no splinters, no protruding screws. Every spring: re-coat anything that has worn through to bare wood.